

Chuqin (Allen) Geng

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Email: chuqin.geng@mail.mcgill.ca

Address: 9 Delphinium Ave, Richmond Hill, ON L4E 4V1

Phone: (514) 652-1880

Website: allengeng123.github.io

Research interests

Neuro-symbolic AI; mechanistic interpretability; interpretable and reliable learning for graph, vision, and language models; neural network robustness and formal verification.

Education

University of Toronto Toronto, ON, CA
Visiting scholar Jan 2023 – Present
Advisor: Prof. Xujie Si

McGill University, Mila - Quebec AI Institute Montreal, QC, CA
PhD in Computer Science Sep 2021 – Present
Advisor: Prof. Xujie Si, GPA: 4.0/4.0

Georgia Institute of Technology Atlanta, GA, USA
MSc in Computer Science, GPA: 4.0/4.0 Sep 2019 – Apr 2021

University of Reading Reading, Berkshire, UK
MSc in Finance, GPA: 4.0/4.0 Sep 2016 – Dec 2017

University of Toronto Toronto, ON, CA
Honours BSc in Math and Statistics Sep 2011 – Aug 2015

Selected Papers

* denotes equal contribution.

Are We Recovering Mechanisms? Objective-Level Recovery Gaps in Mechanistic Interpretability

Chuqin Geng, Li Zhang, Haolin Ye, Mark Zhang, Luke Zhang, Xujie Si
arXiv preprint, 2026

Rethinking Circuit Evaluation: Do Circuits Explain Model Errors?

Li Zhang*, **Chuqin Geng***, Mark Zhang, Chen Yang, Luke Zhang, Haolin Ye, Xujie Si
arXiv preprint, 2026

VisionLogic: Discovering and Grounding Decision-Relevant Visual Concepts

Chuqin Geng, Yuhe Jiang, Li Zhang, Zhaoyue Wang, Haolin Ye, Mark Zhang, Jingkai Xu, Xujie Si
arXiv preprint, 2026

Beyond Message Passing: A Symbolic Alternative for Expressive and Interpretable Graph Learning

Chuqin Geng, Li Zhang, Haolin Ye, Ziyu Zhao, Yuhe Jiang, Tara Saba, Xinyu Wang, Xujie Si
arXiv preprint, 2026

Neural Proposals, Symbolic Guarantees: Neuro-Symbolic Graph Generation with Hard Constraints

Chuqin Geng, Li Zhang, Mark Zhang, Zhaoyue Wang, Haolin Ye, Xujie Si
NeurIPS 2026

DreamProver: Evolving Transferable Lemma Libraries via a Wake-Sleep Theorem-Proving Agent

Youyuan Zhang*, Jialiang Sun*, Hangrui Bi, **Chuqin Geng**, Wenjie Ma, Zhaoyu Li, Xujie Si
COLM 2026

LogicXGNN: Grounded Logical Rules for Explaining Graph Neural Networks

Chuqin Geng, Ziyu Zhao, Zhaoyue Wang, Haolin Ye, Yuhe Jiang, Xujie Si
ICLR 2026 (Top ~2%)

NEUROLOGIC: From Neural Representations to Interpretable Logic Rules

Chuqin Geng, Anqi Xing, Li Zhang, Ziyu Zhao, Yuhe Jiang, Xujie Si
UCRL Workshop @ ICLR, 2026

Learning Minimal Neural Specifications

Chuqin Geng, Zhaoyue Wang, Haolin Ye, Xujie Si
NeuS 2025 (Oral)

Towards Robust Saliency Maps

Van Nham Le*, **Chuqin Geng***, Xujie Si, Arie Gurfinkel
ACML 2024

TorchProbe: Fuzzing Dynamic Deep Learning Compilers

Qidong Su, **Chuqin Geng**, Gennady Pekhimenko, Xujie Si
accepted to APLAS 2023

Scalar Invariant Networks with Zero Bias

Chuqin Geng, Xiaojie Xu, Haolin Ye, Xujie Si
accepted to NeurReps @ NeurIPS 2023

Towards Reliable Neural Specifications

Chuqin Geng*, Van Nham Le*, Xiaojie Xu, Zhaoyue Wang, Arie Gurfinkel, Xujie Si

accepted to ICML 2023 (Oral)

Identifying Different Student Clusters in Functional Programming Assignments: From Quick Learners to Struggling Students

Chuqin Geng, Wenwen Xu, Yingjie Xu, Brigitte Pientka, Xujie Si

accepted to SIGCSE 2023 TS

Novice Type Error Diagnosis with Natural Language Models

Chuqin Geng, Haolin Ye, Yixuan Li, Tianyu Han, Brigitte Pientka, Xujie Si

accepted to APLAS 2022

SAT-DIFF: A Tree Diffing Framework Using SAT Solver

Chuqin Geng, Haolin Ye, Yihan Zhang, Brigitte Pientka, Xujie Si

preprint, arxiv.org/abs/2404.04731

Can ChatGPT Pass An Introductory Level Functional Language Programming Course?

Chuqin Geng, Yihan Zhang, Brigitte Pientka, Xujie Si

preprint, arxiv.org/abs/2305.02230

Selected awards

Apple Scholars nomination (McGill University)	2023
Computer Science Top-up Award (McGill University)	2023
Grad Excellence Award (McGill University)	2021–2023
Academic Distinction Award (University of Reading)	2017
Second prize in mathematics competition (Shaanxi province, China)	2009

Research Experience

Explainable AI and Mechanistic Interpretability

Advisor: Prof. Xujie Si (University of Toronto) *May 2025 – Present*

Developed a novel framework to extract interpretable logic rules and circuits from deep learning models, including GNNs and CNNs.

Leveraged mechanistic interpretability to diagnose model failure modes and applied symbolic fixes to provide formal guarantees and improve robustness.

Extended these neuro-symbolic methods to large language models and agentic theorem proving.

Neural network robustness and verification

Advisor: Prof. Xujie Si (University of Toronto) *Sep 2022 – May 2025*

Proposed new specifications for neural network verification. Explored novel methods to improve models' robustness and fairness.

Disproof of a conjecture in biometric security optimization

Mentors: Prof. Steven Rayan (University of Toronto) Jan 2015 – Aug 2015
Disproved a conjecture regarding optimal solutions for biometric privacy-security trade-offs, providing new bounds for secure system optimization.

Teaching experience

Head teaching assistant, McGill University Winter 2022
COMP 302: Programming Languages and Paradigms
Conducted weekly office hours and tutorials, designed and graded exams, developed auto-graders for assignments, implemented mutation testing, and utilized Moss for plagiarism detection.

Industry experience

FITFI Inc. Toronto, CA
Senior Data Scientist Jan 2018 – Sep 2019
Invented the patent “*System and method for automatically detecting and monitoring use of exercise equipment*”. Led algorithm team, secured demo opportunity at 2019 Collision Conference, and helped raise over 2 million CAD funding.

SHAREWIN SOFTWARE Beijing, China
Algorithm Engineer Sep 2015 – Sep 2016
Designed fault extraction algorithm with preprocessing, filtering, and ant tracking for accurate fault surface detection in 3D-seismic volumes.

Talks

Learning Minimal Neural Specifications May 2025
NeuS 2025

Towards Reliable Neural Specifications
ICML 2023 Aug 2023

A study on student performance clusters Mar 2023
SIGCSE 2023, SPLICE Workshop

Novice Type Error Diagnosis with Natural Language Models Dec 2022
APLAS 2022

Service

I contribute to academic service through conference reviewing and student selection:

- **Machine Learning:** ICML (2023–2026), ICLR (2024–2027), NeurIPS (2023–2026), AAAI (2024–2027)
- **Computer Vision:** CVPR (2024, 2025), ECCV 2026
- **HCI & Education:** SIGCSE 2023, CHI 2023
- **Supervision Evaluation Committee Member:** Mila Supervision Request Process 2027

Mentoring

Li Zhang, Ziyu Zhao, Haolin Ye, Yihan Zhang, Zhaoyue Wang, Xiaojie Xu